Qun Niu

Beijing Institute of Technology Ph.D. Candidate in M.E. (Expected graduation: Dec. 2026)
Target: Industry R&D Positions / Postdoctoral Research Opportunities
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Profile

Ph.D. candidate in Mechanical Engineering at Beijing Institute of Technology, specializing in robotic perception, motion planning, and robust grasp synthesis for complex, unstructured manipulation scenarios. Research integrates 3D vision, geometric reasoning, analytical kinematics, hand-eye/tool calibration, and language-guided grasping into closed-loop robotic systems. Complements analytical kinematics and geometric priors with data-driven paradigms, actively investigating Vision-Language-Action (VLA) models and reinforcement learning for embodied manipulation. Solid engineering background in C/C++, Python, PyTorch, and ROS, with extensive experience deploying dual-arm, mobile, and industrial robotic systems.

Education

Beijing Institute of Technology — Sep. 2020 – Present
Mechanical Engineering — Ph.D.

Beijing University of Technology — Sep. 2017 – Jun. 2020
Mechanical Engineering — M.S.

Hebei University of Science and Technology — Sep. 2013 – Jun. 2017
Process Equipment and Control Engineering — B.S.

Research Interests

- Robot perception: scene segmentation, point-cloud classification, pose estimation, geometric fitting, and 3D vision system integration.
- Kinematics, motion planning, and calibration: inverse kinematics, path planning, collision detection, hand-eye calibration, and tool coordinate calibration.
- Vision-guided grasping: hand-eye calibration, tool coordinate calibration, grasp pose generation, and grasp filtering.
- Language-guided grasping: combining multimodal object localization, geometric reasoning, and task constraints for grasp planning.

Publications

- **Niu, Qun**, Zhang, Chuanlin, Zhang, Tianyu, Zhao, Jieliang, Fu, Tie, and Chen, Xuemei. Language-Guided Robot Grasping Based on Basic Geometric Shape Fitting. *Advanced Intelligent Systems*, e202501276, 2025.
- **Niu, Qun**, Zhao, Jieliang, Liang, Lulu, Xing, Jin, Li, Hongkun, and Wang, Ziru. A General Framework for the Analytical Inverse Kinematics Solution of Industrial Robots. *Proceedings of the Institution of Mechanical Engineers, Part C: Journal of Mechanical Engineering Science*, 239(12):4499–4511, 2025.
- **Niu, Qun**, Wang, Ziru, Li, Hongkun, and Zhao, Jieliang. A Parameters Optimization Framework for Pose Estimation Algorithm Based on Point Cloud. *Journal of Physics: Conference Series*, 2746(1):012039, 2024.
- **Niu, Qun**, Zhao, Jieliang, Liang, Lulu, and Xing, Jin. Dynamic Vertical Climbing Mechanism of Chinese Dragon-Li Cats Based on the Linear Inverted Pendulum Model. *Journal of Bionic Engineering*, 20(1):136–145, 2023.
- **Niu, Qun**, et al. A Simultaneous Hand-Eye and Tool Coordinate System Calibration Framework for Robotic Grasping. *Intelligent Service Robotics* (under review).

- **Niu, Qun**, et al. Enhancing Scene Perception: Shape Fitting with Geometric Primitives. *Intelligent Service Robotics* (under review).

Contribution Overview

- Proposed a general analytical inverse kinematics framework that enables industrial robots with the same configuration to share one D-H-parameter-based solver.
- Introduced a joint calibration model for hand-eye transformation and tool coordinates to reduce error propagation in robotic grasping.
- Built a parameter optimization framework for point-cloud pose estimation to reduce repetitive manual tuning for different objects.
- Developed a scene understanding pipeline that combines segmentation, point-cloud classification, and geometric primitive fitting for interpretable perception.
- Proposed a language-guided model-free grasping method that integrates multimodal localization, primitive fitting, and task-oriented grasp selection.
- Analyzed dynamic vertical climbing in domestic cats with a linear inverted pendulum model and clarified the key mechanism behind dynamic wall traversal without static adhesion.

Patents

- Zhao, Jieliang; Niu, Qun; et al. Adjustable dual-slider ramp experimental platform with tunable height and angle. Granted invention patent, CN202110431126.4.
- Niu, Qun; Zhao, Jieliang; et al. Joint-angle generation method for robots and robot system. Granted invention patent, CN202310966039.8.
- Niu, Qun; Zhao, Jieliang; et al. Vision-guided robotic system and robot calibration method. Granted invention patent, CN202410192128.6.
- Niu, Qun; Zhao, Jieliang; et al. Pose-estimation model configuration method and pose-estimation method. Patent application (under substantive examination), CN202311698845.8.
- Co-authored 9 collaborative papers on quadruped and wall-climbing robots, along with 6 additional granted or pending robotics patents.

Academic Projects

Humanoid Motion Coordination for a Dual-Arm Robot Sep. 2017 – Jun. 2020
Core member Academic project
Designed a cable-driven seven-DOF manipulator, established its D-H-based kinematic model, proposed an inverse kinematics algorithm for humanoid motion, and developed Cartesian-space and joint-space planning methods for shaft-hole assembly tasks.

Biomimetic Quadruped Robot with Integrated Joint Actuation Sep. 2020 – Aug. 2022
Core member Academic project
Developed insect-inspired integrated joint actuators and investigated feline vertical dynamic climbing mechanisms. Acquired motion sequences via high-speed imaging, analyzed contact dynamics and impact dissipation via the SLIP model, and designed an elastic actuator prototype featuring active compliance and passive restraint.

Object Perception and Motion Planning for Robotic Grasping Aug. 2022 – Aug. 2024
Core member Academic project
Led the algorithmic pipeline for unstructured bin picking and cluttered grasping: developed multi-DOF robot kinematics, collision-free trajectory planning, robust hand-eye calibration, point-cloud pose estimation, and contact-stable grasp pose generation with geometric filtering.

Internship Experience

Beijing Modi Technology Co., Ltd. Apr. 2018 – Apr. 2019
R&D Department Internship

Developed forward and inverse kinematics algorithms, joint-space planning, and Cartesian-space planning for a biomimetic four-axis manipulator based on lever mechanisms. Implemented human-machine interfaces and coordinated drive control for a multi-mode cell-culture bioreactor.

Institute of Automation, Chinese Academy of Sciences Apr. 2019 – May 2020
Research Center for Digital Content Technology and Service Internship
Developed adaptive polishing-path planning on quadric surfaces and closed-loop force-controlled polishing head programs for industrial robotic finishing, ensuring on-site delivery. Implemented multi-axis synchronized PLC control and PID bolt pre-tightening control for an automatic screw-locking machine; assisted in modular waterproof mechanical design for a snake robot.

Beijing Transfertech Co., Ltd. Aug. 2022 – Aug. 2024
R&D Department Internship
Developed and maintained robot kinematics, motion planning, and collision detection algorithms. Contributed to grasp pose generation and filtering, exploratory research on model-free grasping, parameter optimization for pose estimation, hand-eye calibration, and a Blazor-based 3D scene configuration module for environment visualization.

Beijing Galbot Co., Ltd. Aug. 2025 – Feb. 2026
Algorithm Engineering Internship
Contributed to embodied intelligence algorithm development for indoor mobile manipulation platforms across perception, decision-making, and control. Developed autonomous grasping and mobile manipulation skills for the Galbot G1 dual-arm humanoid robot in retail capsule scenarios, combining geometric heuristics with learning-based policies.

Skills

- Programming and tools: C/C++, Python, PyTorch, ROS / ROS 2, Blazor, Linux, Git, MATLAB.
- Robot algorithms: kinematics, analytical inverse kinematics, path planning, collision detection, grasp pose generation, and grasp filtering.
- Vision and perception: hand-eye calibration, pose estimation, point-cloud processing (PCL/Open3D), scene segmentation, geometric fitting, and 3D vision systems.
- Embodied AI and learning paradigms: Vision-Language-Action (VLA) models, deep reinforcement learning (RL), and imitation learning in robotic manipulation.

Personal Strengths

- Dedicated to rigorous technical investigation with sustained problem-solving drive.
- Proactive in tackling challenging problems and venturing into emerging paradigms.
- Highly adaptable across diverse research themes, industrial domains, and engineering software stacks.

Honors and Awards

- Outstanding Graduate Student Award, Beijing University of Technology.
- First Prize for Graduate Academic Excellence.
- Graduate Research Excellence Award.
- Second Prize, Hebei Mechanical Design Innovation Competition.
- Athletic awards: 1 municipal-level award and 5 university-level basketball awards.